İlaç Yeniden Konumlandırma ve Toksisite — Teknik Rapor

Profil: smoke; seed: 42.
Amaç: bilgi çizgesi bağlantı skorunu moleküler assay riskiyle birlikte incelemek.
Kısa profil sonuçları çalışma doğrulamasıdır; tam deney sonucu olarak yorumlanamaz.

Veri ve lisans
PrimeKG DOI: 10.7910/DVN/IXA7BM. Kaynak manifesti reports/data_manifest.json içindedir.
PrimeKG yazılımının MIT lisansı, DrugBank dahil kaynak verilerin lisansını kapsamaz.
Tox21: DeepChem MoleculeNet dağıtımı; 12 in vitro assay. Kaynak URL ve hash manifesttedir.

Yöntem
KG: type:source_id kimlikleri; aynı çift arasındaki tüm yön/ilişkiler tek splitte.
TransE/RotatE validation MRR ile seçilir; filtered test tüm bilinen pozitifleri filtreler.
Tox21: canonical isomeric SMILES, duplicate çelişkiler NaN; eksikler negatif sayılmaz.
Morgan radius=2, 1024 bit. Fragment/tautomer standardizasyonu scaffold grup kimliğinde kullanılır.
Scaffold grupları bölünmez; acyclic moleküller tek grup. Oranlar yaklaşık %80/10/10.
XGBoost: görev başına class weight, validation AP seçimi. MLP: 1024/512/256/12, maskeli BCE.
MLP validation macro ROC-AUC ile erken durdurulur. Test model seçiminde kullanılmaz.
PR-AUC average precision; %95 bootstrap CI yalnızca iki sınıflı çekimlerde hesaplanır.

Gerçekleşen deneyler

Kaynak manifesti

```
{
  "retrieved_unix": 1791460308.431595,
  "primekg_doi": "doi:10.7910/DVN/IXA7BM",
  "origin": "pinned_community_mirror",
  "official_access_error": "403 Client Error: Forbidden for url:
https://dataverse.harvard.edu/api/datasets/:persistentId/?persistentId=doi%3A10.7910%2FDVN%2FIXA7BM",
  "primekg_version": "mirror commit 03de2c6613831c66203fb8af8ac7f3fdae6b1887; official version unverified",
  "license_metadata": "See PrimeKG original source licenses; mirror does not grant new rights.",
  "sources": [
    {
      "url": "https://huggingface.co/datasets/hieupth/primekg/resolve/03de2c6613831c66203fb8af8ac7f3fdae6b18
87/org/kg.csv",
      "file": "data/raw/primekg/primekg_raw.csv",
      "sha256": "0a5a010fcd88dcec572a573d98c26c1c244a11567092b6161d3ded611cfb5c69"
    },
    {
      "url": "https://deepchemdata.s3-us-west-1.amazonaws.com/datasets/tox21.csv.gz",
      "file": "data/raw/tox21/tox21.csv.gz",
      "sha256": "45d09792492ce049039dd24aa27b07fc79ce20c573187d4d90bcd178c0c0d360"
    }
  ],
  "license_note": "PrimeKG software MIT != underlying source licenses; DrugBank is restricted."
}
```


Resmî PrimeKG deposu OptimusKG güncellemesini bildirir; bu deney sabit PrimeKG MVP baseline kullanır.
Kaynak: https://github.com/mims-harvard/PrimeKG ; OptimusKG şema desteği uygulanmadı.

Otomatik testler

{"name": "pytest", "errors": "0", "failures": "0", "skipped": "0", "tests": "12", "time": "12.506",
"timestamp": "2026-10-08T12:02:56.981587+00:00", "hostname": "TOFAVDDEV-0"}

kg_data_report.json

```
{
  "source_rows": 8100498,
  "source_relations": {
    "protein_protein": 642150,
    "drug_drug": 2672628,
    "contraindication": 61350,
    "drug_protein": 51306,
```

```
    "indication": 18776,
    "off-label use": 5136,
    "disease_phenotype_positive": 300634,
    "phenotype_phenotype": 37472,
    "phenotype_protein": 6660,
    "disease_phenotype_negative": 2386,
    "disease_protein": 160822,
    "drug_effect": 129568,
    "bioprocess_bioprocess": 105772,
    "disease_disease": 64388,
    "molfunc_protein": 139060,
    "cellcomp_protein": 166804,
    "molfunc_molfunc": 27148,
    "cellcomp_cellcomp": 9690,
    "bioprocess_protein": 289610,
    "anatomy_protein_present": 3036406,
    "pathway_protein": 85292,
    "anatomy_anatomy": 28064,
    "pathway_pathway": 5070,
    "exposure_disease": 4608,
    "exposure_exposure": 4140,
    "exposure_bioprocess": 3250,
    "exposure_protein": 2424,
    "exposure_molfunc": 90,
    "exposure_cellcomp": 20,
    "anatomy_protein_absent": 39774
  },
  "source_endpoint_types": {
    "gene/protein": 5262458,
    "drug": 5611392,
    "disease": 682488,
    "effect/phenotype": 514192,
    "biological_process": 504404,
    "molecular_function": 193446,
    "cellular_component": 186204,
    "anatomy": 3132308,
    "exposure": 18672,
    "pathway": 95432
  },
  "missing": {
    "relation": 0,
    "x_id": 0,
    "x_type": 0,
    "x_name": 0,
    "y_id": 0,
    "y_type": 0,
    "y_name": 0
  },
  "entities": 231,
  "triples": 1800,
  "profile": "smoke"
}

### split_report.json
{
  "seed": 42,
  "sizes": {
    "train": 1440,
    "validation": 180,
    "test": 180
  },
  "requested_ratios": [
    0.8,
```

```
    0.1,
    0.1
  ],
  "actual_ratios": [
    0.8,
    0.1,
    0.1
  ],
  "pair_group_overlap": [
    0,
    0,
    0
  ],
  "entity_coverage": true,
  "relation_coverage": true,
  "policy": "All relations and directions for an unordered endpoint pair stay in one split."
}
```


fingerprint_metadata.json


```
{
  "rdkit_version": "2026.03.6",
  "radius": 2,
  "n_bits": 1024,
  "binary": true,
  "include_chirality": false,
  "canonical_smiles": "isomeric, charge/stereo/salt retained",
  "split_identity": "fragment parent + canonical tautomer + achiral Murcko; acyclic together",
  "shape": [
    7822,
    1024
  ],
  "seconds": 115.52872743199987,
  "raw_rows": 7831,
  "clean_rows": 7822,
  "invalid_rows": 8,
  "collapsed_duplicates": 0,
  "invalid_scaffold_rows": 1,
  "conflicting_labels": 0
}
```


scaffold_split_report.json


```
{
  "seed": 42,
  "sizes": {
    "train": 6258,
    "validation": 782,
    "test": 782
  },
  "scaffold_overlap": [
    0,
    0,
    0
  ],
  "requested_ratios": [
    0.8,
    0.1,
    0.1
  ],
  "actual_ratios": [
    0.8000511378164152,
    0.09997443109179238,
    0.09997443109179238
  ],
```

```
  "acyclic_policy": "all empty scaffolds stay together",
  "group_count": 2295
}
```


mapping_report.json


```
{
  "total_drugs": 50,
  "queried_this_run": 32,
  "verified_structures": 36,
  "unmatched": 14,
  "policy": "DrugBank ID must appear in PubChem CID synonyms or explicit curated verification."
}
```


inference_timing.json


```
{
  "seconds": 0.10616749900009381,
  "candidate_count": 30,
  "complete_risk_candidates": 24,
  "cold_model_load_excluded": true
}
```


kge_comparison.csv

model	dimension	lr	split	MRR	Hits@1	Hits@3	Hits@10	validation_MRR
TransE	32	0.005	test	0.019659	0.000000	0.005556	0.030556	0.020319
RotatE	32	0.005	test	0.084608	0.027778	0.058333	0.177778	0.081808

model_comparison.csv

model	ROC-AUC	PR-AUC	F1	Balanced Accuracy	evaluable_assays
MLP	0.744105	0.346993	0.321205	0.683230	12
XGBoost	0.706299	0.301203	0.278990	0.639067	12

toxicity_final_metrics.csv

model	assay	n	positive	status	ROC-AUC	PR-AUC	Precision	Recall	F1	Balanced Accuracy	bootstrap_valid	ROC_CI_low	ROC_CI_high	PR_CI_low	PR_CI_high
XGBoost	NR-AR	720	32	ok	0.840298	0.520615	0.164948	0.500000	0.248062	0.691134	40	0.772489	0.910291	0.380878	0.701803
XGBoost	NR-AR-LBD	638	23	ok	0.842983	0.383724	0.206897	0.521739	0.296296	0.723471	40	0.784081	0.941172	0.253596	0.617637
XGBoost	NR-AhR	626	95	ok	0.780375	0.408206	0.240122	0.831579	0.372642	0.680385	40	0.749103	0.816167	0.364892	0.489483
XGBoost	NR-Aromatase	521	51	ok	0.647184	0.248074	0.156627	0.509804	0.239631	0.605966	40	0.587445	0.727264	0.186149	0.397261
XGBoost	NR-ER	571	76	ok	0.603495	0.276819	0.182320	0.434211	0.256809	0.567610	40	0.543113	0.658403	0.201115	0.363037
XGBoost	NR-ER-LBD	676	32	ok	0.649384	0.347745	0.144737	0.343750	0.203704	0.621409	40	0.555147	0.743339	0.193334	0.449045
XGBoost	NR-PPAR-gamma	590	30	ok	0.666994	0.121058	0.104651	0.300000	0.155172	0.581250	40	0.627056	0.747031	0.070303	0.212113
XGBoost	SR-ARE	489	107	ok	0.682524	0.390018	0.295455	0.728972	0.420485	0.621030	40	0.641166	0.736122	0.333141	0.471774
XGBoost	SR-ATAD5	679	33	ok	0.732456	0.136241	0.128834	0.636364	0.214286	0.708275	40	0.619028	0.804979	0.094831	0.262416
XGBoost	SR-HSE	597	54	ok	0.696815	0.235446	0.200000	0.425926	0.272189	0.628248	40	0.645173	0.780920	0.142496	0.337171
XGBoost	SR-MMP	514	114	ok	0.687478	0.375166	0.346535	0.614035	0.443038	0.642018	40	0.626233	0.741556	0.328165	0.479166
XGBoost	SR-p53	639	60	ok	0.645596	0.171323	0.145631	0.500000	0.225564	0.598014	40	0.561432	0.729921	0.120072	0.231722
MLP	NR-AR	720	32	ok	0.752316	0.507835	0.132867	0.593750	0.217143	0.706759	40	0.676364	0.869708	0.361707	0.694147
MLP	NR-AR-LBD	638	23	ok	0.865465	0.414236	0.250000	0.695652	0.367816	0.808802	40	0.799994	0.939445	0.283938	0.597239
MLP	NR-AhR	626	95	ok	0.822500	0.521748	0.336364	0.778947	0.469841	0.751997					

40 0.794313 0.864737 0.453240 0.592369
MLP NR-Aromatase 521 51 ok 0.741239 0.306515 0.250000 0.627451 0.357542 0.711598
40 0.672036 0.812503 0.226668 0.434522
MLP NR-ER 571 76 ok 0.641414 0.300764 0.211957 0.513158 0.300000 0.610114
40 0.589505 0.703665 0.236727 0.384307
MLP NR-ER-LBD 676 32 ok 0.665906 0.259890 0.125000 0.375000 0.187500 0.622283
40 0.563198 0.754681 0.132682 0.432269
MLP NR-PPAR-gamma 590 30 ok 0.757143 0.170005 0.113402 0.366667 0.173228 0.606548
40 0.710714 0.829808 0.093755 0.315664
MLP SR-ARE 489 107 ok 0.749180 0.457681 0.378947 0.672897 0.484848 0.681998
40 0.701500 0.795480 0.377823 0.533173
MLP SR-ATAD5 679 33 ok 0.800990 0.259508 0.160839 0.696970 0.261364 0.755606
40 0.727746 0.871868 0.170000 0.450798
MLP SR-HSE 597 54 ok 0.689414 0.273247 0.204082 0.370370 0.263158 0.613362
40 0.627207 0.748075 0.181828 0.376435
MLP SR-MMP 514 114 ok 0.762840 0.444405 0.432927 0.622807 0.510791 0.695154
40 0.705464 0.810375 0.390527 0.541029
MLP SR-p53 639 60 ok 0.680858 0.248080 0.172973 0.533333 0.261224 0.634542
40 0.618026 0.749424 0.159460 0.319165

experiment_log.csv
step seconds seed profile status unix_time
download 24.603144 42 smoke completed 1.791460e+09
prepare-kg 49.805778 42 smoke completed 1.791460e+09
map-drugs 31.640207 42 smoke completed 1.791461e+09
prepare-kg 55.156626 42 smoke completed 1.791461e+09
prepare-tox21 115.557885 42 smoke completed 1.791461e+09
train-kg 58.190115 42 smoke completed 1.791461e+09
map-drugs 45.186992 42 smoke completed 1.791461e+09
train-toxicity 49.901899 42 smoke completed 1.791461e+09
infer 4.701651 42 smoke completed 1.791461e+09
report 1.003565 42 smoke completed 1.791461e+09
infer 6.294498 42 smoke completed 1.791461e+09

Sıralama ve açıklanabilirlik
Hard filter: 12 assay tamamlanmış ve maksimum risk eşikten düşük. Bilinmeyenler elenir.
Ağırlıklı skor: w × min-max KG − (1−w) × maksimum assay riski; eksikler skorlanmaz.
Pareto: KG yüksek ve risk düşük hedefleri. Eşit noktalar birlikte korunur.
sensitivity.csv ağırlık 0/.25/.5/.7/.75/1 sonuçlarını kaydeder; varsayılan .7 klinik optimum değildir.
XGBoost native TreeSHAP bit katkıları log-odds birimindedir; hash çakışmaları açıklamayı sınırlar.
Yerel 1–2 hop çizge patikaları gözlenen ilişkileri gösterir; nedensellik kanıtı değildir.

Sınırlılıklar ve tamamlanmamış deneyler
Tam PrimeKG ve uzun full profil bu kısa doğrulamanın yerine geçmez; ayrı yürütülmelidir.
Assay riskleri kalibre edilmemiştir. Tox21 assay aktivitesi klinik toksisite ile eşdeğer değildir.
KG hedeflerinin test metrikleri tüm ilişkilere aittir; drug-disease özel prospektif doğrulama yapılmadı.
RDKit tautomer enumeration sınırları geçerlidir; karmaşık moleküllerde tüm tautomerler tüketilmemiş olabilir.
Kimyasal eşleme DrugBank kimliği doğrulanamayan adaylarda bilinmeyendir; kapsam mapping_report içindedir.
Salt ve stereokimya retained; fingerprint chirality kapalı. Scaffold kimliği achiral tautomer parenttır.
Ablasyon/çoklu seed/prospektif klinik doğrulama yapılmadı. Hedef metrikler başarı garantisi değildir.
UI 3–5 sn hedefi gözlenen çıkarım süresiyle karşılaştırılır; cold load ve 3D süreleri ayrıca değişebilir.

Tekrar üretme
README komutları, configs/*.yaml, requirements.txt ve manifest SHA256 kayıtlarını kullanın.